

DATE	REVISION	LOCATION	COMMENT
4 MAY 2026			INITIAL ISSUE.
10 MAY 2026			HARDWARE VERIFICATION COMPLETE.

FOR A1, THE ORIGINAL DRAWING SPECIFIES LM1458, HOWEVER ALMOST ANY DUAL OP-AMP WILL WORK. THIS IS NOT A CRITICAL COMPONENT. IF YOU CHOOSE TO ONLY USE ONE OUTPUT AS IN THE PRODUCTION TM-7, YOU CAN USE WHATEVER SINGLE OP-AMP.

OR NOT, I'M NOT YOUR MOM OR WHATEVER.

CONSULT THIS
IMPORTANT SHIT:

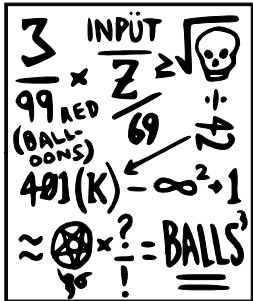


image credit: david c. lovelace
(<https://umop.com/tm7.htm>)

TOTAL STEADY HEATER DRAW ~650mA
IT WILL BE HIGHER AT STARTUP - 7806 MUST BE HEATSINKED ADEQUATELY!
THERE ARE MANY WAYS TO PROVIDE FILAMENT VOLTAGE, THIS IS JUST A SUGGESTION.

ERIC'S ORIGINAL DRAWING SHOWS TWO OUTPUTS, BUT HE MUST HAVE REALIZED THAT TWO OUTPUTS WOULD BE TOO CONFUSING FOR NOT ONLY ALL GUITARISTS, BUT MOST ANY OTHER MUSICIAN. SO HE HAD TO DUMB IT DOWN. THAT'S MY THEORY.

THE STRIPBOARD LAYOUT ASSUMES DUAL OUTPUTS BUT IT WILL WORK WITH JUST ONE OUTPUT AS WELL.

BUILT FROM SCHEMATIC AND VERIFIED AS OF 10 MAY 2026

DISCLAIMER:

PROVIDED WITH NO GUARANTEES, WARRANTIES, OR SUPPORT FROM EITHER TOWMOTOR OR METASONIX.

ORIGINAL CIRCUIT DESIGNED FOR +50VDC, BUT MAY BE OPERATED AS LOW AS 48 VDC OR AS HIGH AS 200 VDC. THE SOURCE, TOPOLOGY, QUALITY, AND RELIABILITY OF THIS VOLTAGE IS THE RESPONSIBILITY OF THE BUILDER. IF THE BUILDER DOESN'T KNOW HOW TO SAFELY GET THAT VOLTAGE, THE BUILDER IS STRONGLY ADVISED TO RECONSIDER CONSTRUCTING THIS CIRCUIT.

MADE WITH DRAWINGS GRACIOUSLY PROVIDED BY ERIC BARBOUR, OWNER OF METASONIX.
DRAWN BY TOWMOTOR ON MODWIGGLER (NOT ASSOCIATED WITH METASONIX)

TM-7 SCROTUM SMASHER.

Size: USLegal Date: 2026-05-10
KiCad E.D.A. 10.0.1

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Id: 1/1